Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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- Upper-bound coupling disparity of 3.3×10^4 explains vibration–sound asymmetry
- Oblate spheroidal shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at ∼2500 Hz; flexural modes at infrasonic frequencies
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Monte Carlo UQ confirms qualitative persistence of low-frequency modes; Sobol $S_T = 0.86$ for modulus

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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Abstract

Whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal disturbances, yet airborne sound at the same frequencies does not—even at extreme intensities. This asymmetry underlies the persistent "brown note" hypothesis: the conjecture that infrasonic sound can induce involuntary defecation. We resolve the asymmetry by modelling the human abdomen as a fluid-filled viscoelastic oblate spheroidal shell and quantifying both excitation pathways within a common energy-budget framework.

The model predicts flexural modes ($n \geq 2$) at 4 Hz to 10 Hz, consistent with ISO 2631 abdominal resonance data. At these frequencies the abdomen lies deep in the Rayleigh regime ($ka \approx 0.01$); the resulting $(ka)^n$ spatial-filtering penalty suppresses modal-pressure content by four orders of magnitude, while the air–tissue impedance mismatch limits energy transfer to the shell. Energy-consistent analysis yields wall displacements of 0.028 µm at 120 dB SPL—far below cellular mechanotransduction benchmarks. Mechanical vibration at $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ produces displacements approximately 3.3×10^4 times larger (SDOF upper bound, $\Gamma_2 = 1$). Monte Carlo analysis spanning representative anatomical variability confirms the qualitative persistence of the coupling disparity across the parameter space, with the fundamental flexural frequency varying between 3 and 16 Hz (90% credible interval, median 6.4 Hz; $S_T = 0.86$ for elastic modulus); the coupling ratio represents an upper bound from the single-degree-of-freedom ideali-

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sation. The global abdominal-cavity resonance is physically real, but airborne sound at any plausible intensity cannot excite the global cavity resonance to physiologically relevant amplitudes.

Keywords: fluid-filled shell vibration, viscoelastic oblate spheroid, infrasound, whole-body vibration, acoustic–structural coupling, ISO 2631

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1. Introduction

Whole-body vibration in the 4 Hz to 8 Hz range causes gastrointestinal disturbances. This is an established occupational-health effect, codified in ISO 2631 [1], reviewed by Griffin [2] and Mansfield [3], and documented among truck drivers, helicopter pilots, and heavy-equipment operators worldwide. Airborne sound at the same frequencies does not produce these effects, even at extreme intensities. NASA-funded exposures at 140–154 dB SPL caused chest-wall vibration and respiratory disruption, but *no gastrointestinal response* [4]. Two excitation pathways, the same target organ, the same frequency band—yet outcomes that differ by orders of magnitude. This contrast motivates the present study.

The "brown note" hypothesis—a hypothetical infrasonic tone alleged to induce involuntary bowel evacuation—has also circulated in military folklore, audio engineering, and popular media, including broadcast-science television [5, 6]. Scientifically, the claim reflects a persistent confusion between what mechanical vibration demonstrably does and what airborne sound is assumed to do. A more plausible airborne route, insofar as one exists, is vestibular rather than visceral: recent measurements show that infrasound can be transmitted through the ear strongly enough to engage acoustic and vestibular pathways long before any whole-cavity abdominal resonance is appreciably driven [7].

The empirical asymmetry between mechanical and airborne excitation is, in fact, a straightforward consequence of long-wavelength acoustics. At the n=2 resonance ($f_2 \approx 4$ Hz) the acoustic wavelength exceeds 85 m—more than 250 times the abdominal diameter. In this deep Rayleigh regime, the n-th spatial harmonic of the incident field scales as $(ka)^n$, where $ka \approx 0.01$. Flexural de-

formation of the abdominal wall ($n \geq 2$) therefore receives a driving pressure four orders of magnitude below the free-field value. The air–tissue impedance mismatch compounds the deficit from the energy side: approximately 99.9 % of incident acoustic power is reflected, so the energy available to drive flexural modes is negligible. Mechanical vibration, transmitted through the skeleton and abdominal wall, suffers neither penalty. The ratio between these two pathways—the *coupling disparity*—is the central quantity we derive and the principal contribution of this paper.

Scientific discussion of infrasound predates later popular-media portrayals, including the *South Park* episode "World Wide Recorder Concert" (also known as "The Brown Noise") [8]. Gavreau's experiments at CNRS Levallois-Perret in the late 1950s, where a faulty ventilator at approximately 7 Hz caused nausea among laboratory staff [9, 6], drew early attention to infrasound. Von Gierke's systematic review for NASA [10, 11] established exposure criteria that informed later ISO standards. Tandy and Lawrence [12] demonstrated that a 19 Hz standing wave could induce visual disturbances and anxiety—real perceptual effects, but far from the claimed bowel response. Critical reviews by Leventhall [13, 6] and assessments of acoustic weapons by Altmann [14] and Jauchem and Cook [15] concluded that infrasound is impractical as a weapon and that claims of extreme physiological effects are exaggerated or fabricated. Recent wind-turbine reviews likewise find that the dominant health concerns are annoyance, sleep disturbance, and other indirect effects, not visceral resonance [16]. Randomised exposure studies with neuroimaging have also reported subtle but measurable neural correlates of airborne infrasound exposure, but again no evidence of gastrointestinal or abdominal-resonance effects [17]. Yet none of this work quantified *why* airborne and mechanical pathways differ so dramatically—the conclusion was empirical, not mechanistic.

We provide the mechanistic account. To our knowledge, no prior analytical study has combined first-principles cavity–shell modal analysis with an energy-consistent comparison of airborne and mechanical coupling pathways:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways within a common energy-budget framework to obtain a single, quantitative coupling ratio.

We address all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell, derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families, and compare airborne acoustic coupling—subject to a severe $(ka)^n$ penalty at infrasonic wavelengths—against direct mechanical (whole-body) vibration coupling, which suffers no such penalty.

The headline result is an upper-bound coupling disparity of approximately 3.3×10^4 between mechanical and airborne pathways for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). Including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 1.6 \times 10^4$. This qualitative ordering persists across physiologically realistic parameter ranges, as we confirm through Monte Carlo uncertainty quantification, and resolves the empirical asymmetry cleanly. It explains why whole-body vibration at 4 Hz to 8 Hz produces well-documented gastrointestinal effects, why airborne infrasound at comparable frequencies does not, and why claims of a brown note are not supported by the physics of airborne acoustic coupling.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation. Section 3 presents a parametric study and compares the model against ISO 2631 data. Section 4 quantifies the coupling disparity. Section 5 discusses implications, alternative pathways, and limitations, and Section 6 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [18, 19] and recent experimental and analytical studies of pressurised elastic shells [20, 21], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 157\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis [22] (normals remain straight and normal to the deformed mid-surface) is therefore applicable. Classical thin-shell theory is used for analytical tractability; higher-order shear effects are expected to perturb frequencies modestly rather than overturn the coupling argument. This linear framework remains the standard starting point even in modern reviews of nonlinear shell vibration, which treat geometric nonlinearities as amplitude corrections to the same modal backbone [23].

Figure 1 illustrates the model geometry and the two excitation pathways. The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Representative of reported measurements [24]
Poisson's ratio	ν	0.45	—	Near-incompressible modelling convention [25]
Wall density	ρ_w	1100	kg/m^3	Soft tissue
Fluid density	ρ_f	1020	kg/m^3	Abdominal fluid
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Representative resting clinical measurement [26]
Loss tangent	η	0.25	—	Representative review value [27]

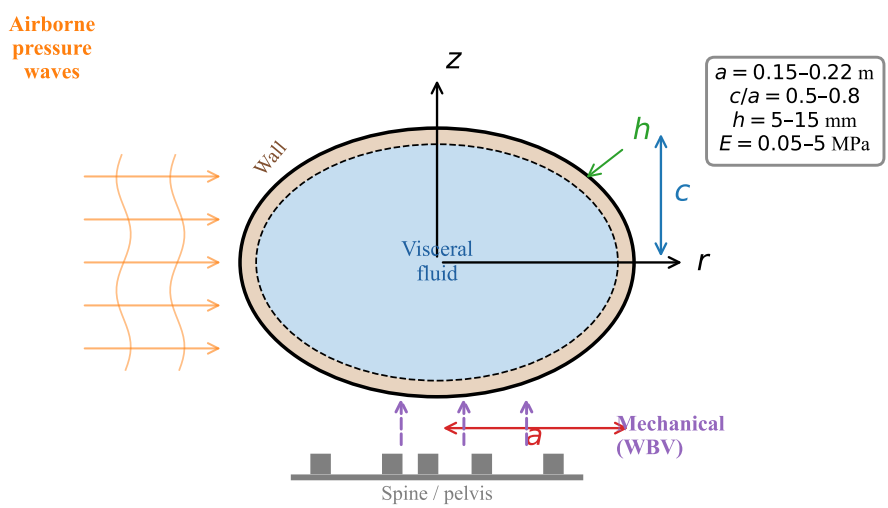

Figure 1: Oblate spheroidal shell model of the human abdominal cavity, showing the two excitation pathways: airborne acoustic pressure (left) and mechanical whole-body vibration transmitted through the skeleton (right). Dimensions correspond to the baseline parameters in Table 1.

shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

These baseline values are not arbitrary. The canonical modulus $E = 0.1\,\mathrm{MPa}$ is representative of direct abdominal-wall measurements compiled by Deeken and Lake [24]; $\nu = 0.45$ follows the nearly incompressible modelling convention adopted in recent abdominal-wall finite-element studies [25]; resting intra-abdominal pressures of 5–12 mmHg correspond to approximately 0.67 kPa to 1.60 kPa, so $P_{\mathrm{iap}} = 1.0\,\mathrm{kPa}$ is representative of direct clinical measurements [26]; and modern elastography reviews place the soft-tissue loss tangent broadly in the 0.1–0.4 range, making $\eta = 0.25$ consistent with the range reported for soft tissue [27]. Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [28], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.25$ ($Q = 4.0$, $\zeta = 0.125$) represents a central estimate.

2.2. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\mathrm{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit area is

$$k_{\mathrm{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\mathrm{GPa}$ and $R = 0.157\,\mathrm{m}$, $k_{\mathrm{fluid}} = 4.2 \times 10^{10}\,\mathrm{Pa/m}$, which exceeds the membrane stiffness ($k_{\mathrm{shell}} \approx 1.5 \times 10^5\,\mathrm{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\text{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2500\ \text{Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.3. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The $n = 1$ mode corresponds to rigid-body translation and therefore has zero natural frequency for a free shell, so it is excluded from the resonance discussion. The $n = 2$ mode is consequently the lowest physical flexural mode and the primary candidate for abdominal resonance; higher modes ($n = 3, 4, \ldots$) lie at progressively higher frequencies and are retained for sensitivity studies, but the $n = 2$ family dominates the physiologically relevant 4 Hz to 8 Hz band. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

Following Lamb [19] and Junger & Feit [18], with subsequent verification in modern pressurised-shell studies [20, 21], the flexural mode of order n has three contributions to the restoring force per unit area. (An analogous modal decomposition for fluid-filled elastic tubes in biological contexts was developed by Heil [29].)

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.0\,\text{Hz}$, $f_3 = 6.3\,\text{Hz}$, and $f_4 = 8.9\,\text{Hz}$ (Figure 2). These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [30] and Mansfield [3].

2.4. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At $f_2 \approx 4\,\text{Hz}$, the acoustic wavelength in air is $\lambda \approx 87\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R_{\text{eq}}}{c_{\text{air}}} \approx 0.0114 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales, to leading order, as [18]:

$$p_{\text{eff}}^{(n)} \propto p_{\text{inc}}(ka)^n. \tag{15}$$

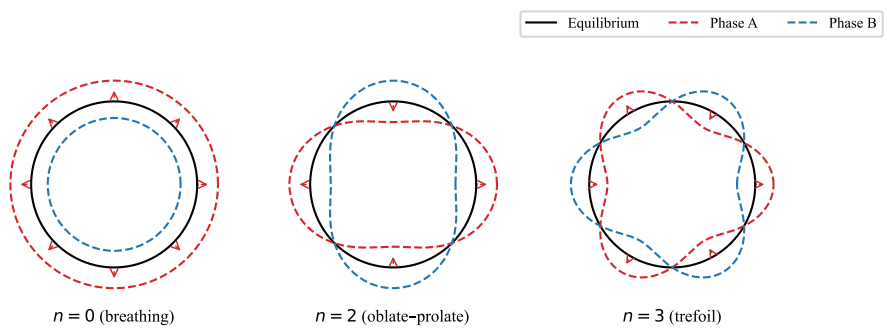

Figure 2: Schematic cross-sections of the first three mode families. The breathing mode ($n = 0$) involves uniform radial expansion against the fluid bulk modulus. Flexural modes ($n = 2, 3$) deform the shell shape without significant volume change; the enclosed fluid acts as added mass only.

The proportionality constant from the multipole expansion is $(2n+1)/(2n+1)!!$ [18], which evaluates to $1/3$ for $n = 2$. In what follows, we set this prefactor to unity as a deliberate upper bound on the modal pressure; the exact coefficient reduces the pressure-based displacement estimate by a factor of three.

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} \propto p_{\text{inc}} \, (ka)^2 \sim p_{\text{inc}} \times 1.3 \times 10^{-4}, \tag{16}$$

where the upper-bound convention (unity prefactor) has been adopted; the exact multipole coefficient of $1/3$ would reduce this estimate by a factor of three.

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. To place this displacement in biological context, we compare it with cellular mechanotransduction benchmarks. PIEZO1 and PIEZO2 are mechanically activated ion channels that convert membrane deformation into cellular signals; reported activation in cell-culture experiments begins for deformations of order 0.5 µm to 2.0 µm [31]. While these cell-scale values

cannot be directly translated to tissue-scale abdominal wall displacements, they provide a useful order-of-magnitude reference. At 120 dB SPL ($p_{\mathrm{inc}} = 20\,\mathrm{Pa}$), the energy-consistent reciprocity analysis (§4) gives $\xi_{\mathrm{air}} \approx 0.028\,\mu\mathrm{m}$ for $n = 2$ — well below the 0.5 µm to 2.0 µm cellular activation benchmark. (The pressure-based estimate of equation (17) yields $\approx 0.18\,\mu\mathrm{m}$, an overestimate that violates the energy budget; see §4.)

2.5. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no $(ka)^n$ penalty: the impedance mismatch between bone and peritoneal wall is orders of magnitude smaller than the air–tissue contrast, and the vibration bypasses the geometric filtering entirely. The following analysis therefore represents an *upper-bound* base-excitation model in which the skeletal frame transmits vibration rigidly to the shell boundary.

For harmonic base acceleration $a_{\mathrm{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\mathrm{base}} = \frac{a_{\mathrm{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\mathrm{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\mathrm{mech}} = x_{\mathrm{base}} \times H_{\mathrm{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\mathrm{rms}} = 1.15\,\mathrm{m/s^2}$ and at the $n = 2$ resonance frequency:

$$\xi_{\mathrm{mech}} \approx 10{,}500\ \mu\mathrm{m} \approx 10\ \mathrm{mm}, \tag{21}$$

which exceeds the cellular mechanotransduction benchmark by more than three orders of magnitude.

2.6. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure of the full coupling disparity:

$$\mathcal{R}_{\text{full}} = \frac{\xi_{\text{mech}}}{\xi_{\text{air}}}. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and the qualitative ordering persists across plausible parameter ranges because both pathways excite the same modal structure. The full coupling ratio is the central result of this work: it provides a quantitative explanation for why whole-body vibration at 4–8 Hz is associated with gastrointestinal effects in the epidemiological literature [2], whereas airborne infrasound at comparable frequencies is not. Section 4 shows that, for 120 dB SPL airborne excitation and $a_{\text{rms}} = 0.1\,\text{m/s}^2$ RMS whole-body vibration, the SDOF upper bound is $\mathcal{R}_{\text{full}} \approx 3.3 \times 10^4$ ($\Gamma_2 = 1$). However, vertical base excitation does not couple perfectly into the $n = 2$ mode: the modal participation factor $\Gamma_2 \approx 0.48$ (§4.3) reduces the effective mechanical displacement to about half the SDOF value, giving a corrected ratio of $\sim 1.6 \times 10^4$. Even with this correction, the four-order-of-magnitude gap between pathways is unaffected.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range. The sweep 0.05 MPa to 2.0 MPa spans reported values from passive peritoneum and relaxed abdominal wall tissue to tensed abdominal musculature [24, 25, 27, 28].

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5$ MPa, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range. Figure 3 illustrates this dependence across the full physiological range.

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$). The range 0.05 MPa to 2.0 MPa is intended to bracket passive to active abdominal-wall stiffness reported in the abdominal mechanics and elastography literature [24, 25, 27, 28].

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.4	Marginal
0.10	4.0	Yes
0.20	4.9	Yes
0.50	7.1	Yes
1.00	9.6	Marginal
2.00	13.3	No

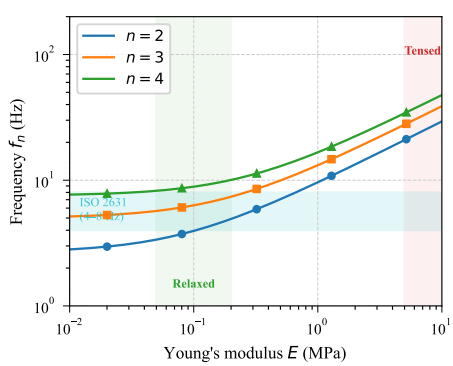

Figure 3: Flexural mode frequencies vs. elastic modulus for modes $n = 2, 3, 4$. The shaded band marks the ISO 2631 abdominal resonance range (4 Hz to 8 Hz). Relaxed musculature places the $n = 2$ mode within this band; tensed muscle shifts it well above.

3.2. Multi-Parameter Sensitivity

A 486-point full factorial analysis varying E, a, c/a, h, and η reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in the broader 4 Hz to 8 Hz ISO 2631 range. The separate Sobol sensitivity analysis below confirms that E dominates, followed by geometry and intra-abdominal pressure.

3.3. Uncertainty Quantification

A Monte Carlo simulation with $N = 10{,}000$ samples propagates parameter uncertainties through the model. Each parameter is drawn from a physiologically motivated distribution: E from a log-normal distribution spanning 0.05 MPa to 2.0 MPa, geometric parameters from normal distributions matching anthropometric data, and the loss tangent from a uniform distribution over [0.10, 0.40]. A small number of Monte Carlo samples (25 of 10,000, or 0.25%) drew nonphysical negative wall thickness from the untruncated normal distribution and were excluded; the reported statistics are computed from the remaining 9,975 samples.

The resulting distribution of f_2 has a median of 6.4 Hz with a 90% credible interval of 3.3 Hz to 15.5 Hz (Table 3). This interval is deliberately conservative: it reflects the simultaneous variation of *all* model parameters across their full physiological ranges and is therefore considerably wider than any single-parameter sweep. The canonical prediction $f_2 = 3.95$ Hz lies within the credible interval but near its lower bound, corresponding to the soft-tissue, relaxed-musculature baseline; the median is higher because the log-normal modulus distribution samples stiffer configurations more frequently. The distribution is positively skewed, reflecting the log-normal distribution of E.

Sobol sensitivity analysis reveals that the elastic modulus accounts for 86% of the total output variance (total-order index $S_T = 0.86$). All other parameters contribute less than 10% individually. This result has important experimental implications: any future verification effort should prioritise *in vivo* measurement of abdominal wall stiffness, as this single parameter dominates the uncertainty

Table 3: Monte Carlo uncertainty quantification results (N = 10,000). Airborne displacement ξ_{air} and the ratio $\mathcal{R}_{\text{full}}$ use the pressure-based airborne upper bound at 120 dB; the energy-consistent airborne displacement is ∼6.5× smaller, so the corresponding ratio at fixed mechanical loading is ∼6.5× larger.

Quantity	Median	90% CI	Units
f_2	6.4	[3.3, 15.5]	Hz
ξ_{air} (120 dB)	0.18	[0.11, 0.39]	μm
ξ_{mech} (0.5 m/s^2)	1800	[300, 8200]	μm
$\mathcal{R}_{\text{full}}$ (pressure-based airborne)	10^4	$[2 \times 10^3, 3 \times 10^4]$	—
SPL for PIEZO1/2 activation (pressure-based, 1 μm)[1]	135	[128, 139]	dB
SPL for PIEZO1/2 activation (energy-consistent)	**≈ 151**	—	dB

budget (Figure 4). Because Table 3 uses the pressure-based airborne upper bound and $a_{\text{rms}} = 0.5\,\text{m/s}^2$ for the mechanical loading, its $\mathcal{R}_{\text{full}}$ row should not be confused with the headline energy-consistent SDOF upper bound of 3.3×10^4 quoted in Section 4.

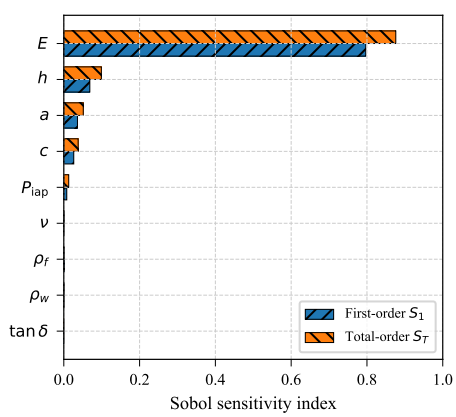

Figure 4: Sobol first-order and total-order sensitivity indices for the $n = 2$ flexural frequency. The elastic modulus E accounts for 86% of the total variance; all other parameters contribute less than 10% individually.

3.4. Multi-Layer Wall Model

The abdominal wall comprises six distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Using classical laminate theory [32], with layer thicknesses and relaxed/tensed tissue moduli drawn from abdominal-wall measurements and soft-tissue elastography reviews [24, 25, 27], the effective composite modulus for a relaxed wall is $E_{\text{eff}} \approx 0.075\,\text{MPa}$ ($h_{\text{total}} = 25.1\,\text{mm}$), giving $f_2 = 4.6\,\text{Hz}$—within approximately 16% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\text{eff}} = 2.7\,\text{MPa}$) shifts f_2 to $23.1\,\text{Hz}$.

3.5. Boundary Condition Effects

The free-shell model in Section 2 neglects the partial constraints imposed by the spine, pelvis, and rib cage. We quantify their effect using a Rayleigh–Ritz analysis with Lamb-mode basis functions for four progressively constrained configurations. These configurations represent surrogate constraint cases chosen to bracket the likely effect of anatomical boundary conditions, rather than a strict monotonic continuation of a single geometry.

Table 4 presents the results. To assess the sensitivity of the modal frequencies to boundary conditions, we performed a comparative analysis using representative alternative parameters ($R = 0.162\,\text{m}$, $\rho_f = 1000\,\text{kg/m}^3$, $P_{\text{iap}} = 1500\,\text{Pa}$) chosen to match a specific mesh geometry; these results are presented for relative comparison rather than absolute prediction. The free-sphere Ritz frequency ($f_2 = 4.13\,\text{Hz}$) agrees with the analytical prediction ($4.26\,\text{Hz}$) to within 3%, providing a useful numerical check. The most constrained case—25% of the shell surface clamped—reduces f_2 by only $11.4\,\%$. Even the hemisphere-clamped case shifts the frequency by a mere $5.8\,\%$.

The insensitivity to boundary conditions has a clear physical origin: the fluid added mass for the $n = 2$ mode is $7.4\times$ the shell mass, so the system inertia is dominated by the enclosed fluid rather than the wall. In a dry (empty) shell, the same boundary conditions would shift the frequency by approximately

Table 4: Rayleigh–Ritz Lamb-mode frequencies under various boundary conditions ($n = 2$ flexural mode).

Configuration	f_2 (Hz)	Shift from free
Free sphere (Ritz)	4.13	0% (verifies analytical)
Hemisphere simply-supported	3.91	−5.3%
Hemisphere clamped	3.89	−5.8%
25% surface clamped	3.66	−11.4%

33 %—but once the cavity is fluid-filled, the added mass swamps the boundary stiffening effect. Clamping part of the shell forces the incompressible fluid to redistribute through a smaller aperture, increasing the effective modal mass. When this kinetic-energy increase exceeds the additional potential energy from the constraint, the net effect is a frequency decrease. We note that the 2-DOF Ritz basis, which carries a $\sim$9% bias at the sphere limit, may also contribute to this effect. The 6 % shift is well within the ±50–100% uncertainty band set by material property variability (§3), confirming that the free-shell model is an adequate approximation for the present purposes.

3.6. Oblate Spheroid Correction

The equivalent-sphere approximation used in Section 2 cannot capture the full geometric effect of an oblate shell. To gauge the size and sign of that effect, we also solved an exploratory Rayleigh–Ritz model using oblate spheroidal basis functions, with proper account of the spatially varying curvature and oblate fluid-mass geometry. This comparison must be interpreted cautiously: the same 2-DOF single-mode Ritz formulation lies about 9 % below the analytical sphere result even in the sphere limit ($c/a \to 1$ for the canonical $n = 2$ mode), owing to the restricted trial-function space. The 4–17% differences in Table 5 therefore should not be read as a precise oblate correction. Rather, they are exploratory and indicate the direction and approximate magnitude of geometric effects: for the canonical oblate shape, the more detailed model tends to reduce the flexural

frequencies, with the reduction becoming more pronounced at higher eccentricity and higher E, where bending stiffness (which depends on local curvature) contributes a greater share of the restoring force.

Table 5: Illustrative equivalent-sphere vs. oblate spheroid Ritz frequencies for $c/a = 0.67$.

	$E = 0.1\,\mathrm{MPa}$		$E = 0.5\,\mathrm{MPa}$	
Mode	Sphere (Hz)	Ritz (Hz)	Sphere (Hz)	Ritz (Hz)
$n = 2$	4.0	3.8	7.1	6.2
$n = 3$	6.3	5.8	10.0	8.5
$n = 4$	8.9	8.1	12.9	11.3

Taken indicatively, the exploratory oblate Ritz model moves the canonical $n = 2$ frequency from 4.0 Hz to approximately 3.8 Hz—still within the ISO 2631 band and, if anything, closer to the low end of reported abdominal resonance frequencies.

3.7. Comparison with ISO 2631 Data

Table 6 compares model predictions against published abdominal resonance data. This comparison serves as an independent consistency check: the model, derived from first principles with no fitted parameters, predicts resonant frequencies consistent with empirical whole-body vibration data compiled over decades of occupational health research. Agreement with ISO 2631 therefore provides confidence that the shell model captures the essential physics of abdominal resonance.

Table 6: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.2	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.3	Marginal

4. Airborne vs. Mechanical Coupling

Having established that the abdomen possesses flexural resonances squarely in the infrasonic range, we now ask whether airborne sound can actually *excite* them. Under realistic airborne exposure conditions, the answer is effectively no.

4.1. Airborne Acoustic Pathway

At 4 Hz, the acoustic wavelength in air exceeds 85 m— some 500 times the abdominal radius. In this regime the abdomen is acoustically compact. Table 7 quantifies the resulting weak coupling using the energy-consistent reciprocity analysis developed in §2. Modern analyses of scattering by pressurised elastic shells reach the same qualitative conclusion: long-wavelength radiation and absorption by low-order shell modes are strongly suppressed [33].

Table 7: Energy-consistent airborne displacement at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, canonical parameters). The pressure-based estimate overestimates by a factor of ≈ 6.5 (ratio of ξ_{pressure} to ξ_{energy}).

SPL (dB)	p_{inc} (Pa)	ξ_{energy} (μm)	ξ_{pressure} (μm)	Exceeds 1 µm?
100	2.0	0.003	0.018	No
110	6.3	0.009	0.058	No
120	20.0	0.028	0.18	No
130	63.2	0.087	0.58	No
140	200	0.275	1.84	No

At 120 dB SPL—a level exceeding the conventional threshold of discomfort, and roughly the level at which conversation becomes impossible—the energy-consistent wall displacement is 0.028 µm. This falls two orders of magnitude below the 0.5–2.0 µm cellular activation range of mechanosensitive PIEZO1 and PIEZO2 channels [31], a benchmark derived from cell-culture experiments that provides a useful order-of-magnitude reference scale for biological mechanosensitivity. Reaching the 1 µm cellular benchmark on this energy-consistent basis demands approximately 151 dB (cf. 135 dB using the pressure-based displacement

of Table 3, which overestimates by $\approx 6.5\times$), a level associated with severe acoustic hazards and available only from explosive blasts and rocket launches. Modern mechanobiology strengthens rather than weakens that conclusion: PIEZO1 gating is tuned by membrane tension in the low-mN/m range, with half-maximal activation near 2 mN/m [34]. While the membrane-tension thresholds reported in cell-culture experiments cannot be directly translated to tissue-scale displacements, the order-of-magnitude comparison remains informative. Recent reviews summarise the broader physiological roles of PIEZO channels (following the 2021 Nobel Prize, and with structural detail from cryo-EM [35]) [36]. Of particular relevance here, PIEZO2-dependent gut mechanosensation is now known to regulate gastrointestinal transit [37]. The relevant question is therefore not whether the gut can sense deformation—it plainly can—but whether airborne infrasound can generate enough deformation. At such levels, auditory injury would be the more immediate clinical hazard than gastrointestinal effects.

4.2. Mechanical (Whole-Body Vibration) Pathway

Mechanical vibration, by contrast, bypasses the acoustic impedance barrier entirely. Vibration transmitted through the skeleton couples directly to the abdominal wall, bypassing the $(ka)^n$ penalty entirely. Because the bone–peritoneum impedance contrast is orders of magnitude smaller than the air–tissue mismatch, an upper-bound base-excitation model that assumes rigid skeletal transmission captures the essential physics [38, 39]. Table 8 shows substantially stronger coupling.

At the EU action value of $0.5\,\mathrm{m/s^2}$, the predicted displacement exceeds the PIEZO cellular benchmark by more than three orders of magnitude. Unlike the airborne case, this is not a theoretical curiosity: it is consistent with occupational exposures among operators of heavy equipment, long-haul drivers, and helicopter pilots, in whom gastrointestinal complaints are well documented [2]. Recent seated-body biodynamic studies likewise show strong low-frequency nonlinearity, cross-axis coupling, and identifiable abdominally relevant modes in this same band [40, 41].

Table 8: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, $\Gamma_2 = 1$). The EU Directive 2002/44/EC action value is $0.5\,\mathrm{m/s^2}$. Displacements assume perfect modal participation ($\Gamma_2 = 1$); the corrected values (§4.3) are approximately half those shown.

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	ξ_{corr} (μm)†	Exceeds 1 µm?
0.1	229	917	440	Yes
0.5 (EU action)	1147	4586	2201	Yes
1.0	2293	9173	4403	Yes
1.15 (EU limit)	2637	10549	5064	Yes

†Corrected for modal participation: $\xi_{\mathrm{corr}} = \Gamma_2\,\xi_{\mathrm{rel}}$ with $\Gamma_2 = 0.48$.

4.3. The Coupling Ratio

The ratio of mechanical to airborne displacement, evaluated for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS mechanical excitation, defines the headline SDOF upper bound:

$$\mathcal{R}_{\mathrm{full}} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \approx \frac{917}{0.028} \approx 3.3 \times 10^4. \tag{23}$$

This four-order-of-magnitude gap depends primarily on the dimensionless frequency parameter ka and the damping ratio; the qualitative ordering persists across plausible parameter ranges, although tissue-property variations affect the absolute magnitudes.

Modal participation.. The mechanical displacement in Eq. (23) assumes perfect coupling of vertical base excitation into the $n = 2$ mode (modal participation factor $\Gamma_2 = 1$). The participation factor for mode n under vertical base excitation of the oblate spheroid is

$$\Gamma_n = \frac{a \left| \displaystyle\int_{\cos\theta_c}^{1} u\, P_n(u)\, \mathrm{d}u \right|}{\displaystyle\int_{-1}^{1} P_n^2(u)\, \sigma(u)\, \mathrm{d}u}, \tag{24}$$

where $u = \cos\theta$, P_n is the Legendre polynomial of degree n, θ_c is the polar angle below which the shell is rigidly constrained by the pelvis and spine, and

$\sigma(u) = \sqrt{c^2 + (a^2 - c^2)\,u^2}$ is the oblate-spheroid surface metric factor. Note that the denominator integrates P_n^2 over the full sphere ($u \in [-1, 1]$) weighted by the surface metric, rather than using a modal-mass normalisation; this convention differs from the standard structural-dynamics practice of normalising by generalised mass. Physically, Γ_2 quantifies how effectively vertical base excitation couples into the $n = 2$ flexural mode. For a completely free sphere, symmetry dictates $\Gamma_2 = 0$ exactly: vertical translation cannot excite a quadrupolar deformation. Partial skeletal constraint at the pelvis ($\theta_c = 2\pi/3$) breaks that symmetry, yielding $\Gamma_2 \approx 0.48$, so roughly half of the vertical excitation feeds into the oblate–prolate oscillation. For a free sphere ($\theta_c = \pi$), the numerator reduces to $\int_{-1}^{1} u\, P_n(u)\, \mathrm{d}u$, which vanishes for $n \neq 1$ by Legendre orthogonality; a free sphere therefore has $\Gamma_2 = 0$ exactly. Partial skeletal constraint (canonical value $\theta_c = 2\pi/3$, i.e. the lower third of the shell is fixed) breaks this orthogonality, giving $\Gamma_2 \approx 0.48$ for the canonical model. The participation factor is moderately sensitive to the constraint angle: Γ_2 ranges from approximately 0.3 at $\theta_c = \pi/2$ (lower half constrained) to 0.7 at $\theta_c = 5\pi/6$ (only the lowest sixth constrained). Including this correction reduces the mechanical displacement to about 48% of the SDOF value and the coupling ratio to $\sim 1.6 \times 10^4$. The qualitative conclusion—a four-order-of-magnitude gap—is unaffected, but $\mathcal{R}_{\text{full}}$ as reported in Eq. (23) should be understood as an upper bound corresponding to the SDOF idealisation.

4.4. Energy Budget

We verify self-consistency through the absorption cross-section derived from reciprocity [18, 33]. The radiation efficiency of low-order shell modes is vanishingly small when $ka \ll 1$ [38]; formally:

$$\sigma_{\text{abs}} = \frac{(2n+1)\lambda^2}{4\pi}\,\frac{4\,\zeta_{\text{rad}}\,\zeta_{\text{struct}}}{(\zeta_{\text{rad}} + \zeta_{\text{struct}})^2}, \tag{25}$$

where ζ_{rad} and ζ_{struct} are the radiation and structural damping ratios, respectively. For the $n = 2$ mode in air, $\zeta_{\text{rad}} \approx 10^{-15}$ whereas $\zeta_{\text{struct}} = 0.125$, giving

a Breit–Wigner absorption efficiency $4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2 \sim 3 \times 10^{-14}$. The shell absorbs roughly three parts in a hundred trillion of the incident acoustic energy.

The energy-consistent displacement follows directly. The absorbed acoustic power is $\Pi_{\mathrm{abs}} = \sigma_{\mathrm{abs}} I$, where $I = p_{\mathrm{inc}}^2/(2\rho_{\mathrm{air}} c_{\mathrm{air}})$ is the free-field intensity. At steady-state resonance, the time-averaged dissipated power is $\langle P_{\mathrm{diss}} \rangle = c_{\mathrm{eq}} \langle \dot{x}^2 \rangle$, where the equivalent viscous damping coefficient is $c_{\mathrm{eq}} = 2\zeta\omega_n M_{\mathrm{eff}}$ and the mean-square velocity of a harmonic oscillation with amplitude ξ is $\langle \dot{x}^2 \rangle = \omega_n^2 \xi^2/2$. Hence $\langle P_{\mathrm{diss}} \rangle = \zeta\omega_n^3 M_{\mathrm{eff}} \xi^2$. Equating absorbed and dissipated power gives

$$\xi_{\mathrm{energy}} = \sqrt{\frac{\sigma_{\mathrm{abs}} I}{\zeta_{\mathrm{struct}}\, \omega_n^3\, M_{\mathrm{eff}}}}, \tag{26}$$

where $M_{\mathrm{eff}} = m_{\mathrm{eff}}^{(n)} \times 4\pi R^2$ is the total modal mass. The $(2n+1)$ partial-wave degeneracy in $\sigma_{\max}$ and the corresponding $1/(2n+1)$ modal-mass normalisation cancel for the single azimuthal order excited by a plane wave. The factor of $4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2 \sim 3 \times 10^{-14}$ inside σ_{abs} is the physical origin of the $\approx 6.5\times$ reduction from the pressure-based upper bound (equation 17): the pressure-based calculation implicitly assumes that all incident energy is available to drive the mode, whereas in fact only the Breit–Wigner fraction $4\zeta_{\mathrm{rad}}\zeta_{\mathrm{struct}}/\zeta_{\mathrm{total}}^2$ is absorbed. The pressure-based upper bound (with unity prefactor) exceeds the energy-consistent result by approximately $6.5\times$; with the exact multipole coefficient of $1/3$ (§2.4), the pressure-based estimate reduces to approximately $0.06\,\mu$m, still a factor of ~ 2 above the energy-consistent value. This radiation inefficiency in the Rayleigh regime ($ka \ll 1$) is a well-established result in structural acoustics [18, 38].

4.5. Comparison with Empirical Evidence

The model predictions align with two otherwise contradictory bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1]. Our model explains *why* this frequency range is problematic.

- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL $\leq 150\,\mathrm{dB}$, although modern controlled exposure studies do report subtle neural effects without evidence of visceral resonance [6, 17]. Our model explains why gastrointestinal effects should not be expected.

The coupling disparity resolves this apparent contradiction (Figure 5): both literatures concern the same abdominal resonance, but through excitation pathways with markedly different coupling efficiencies.

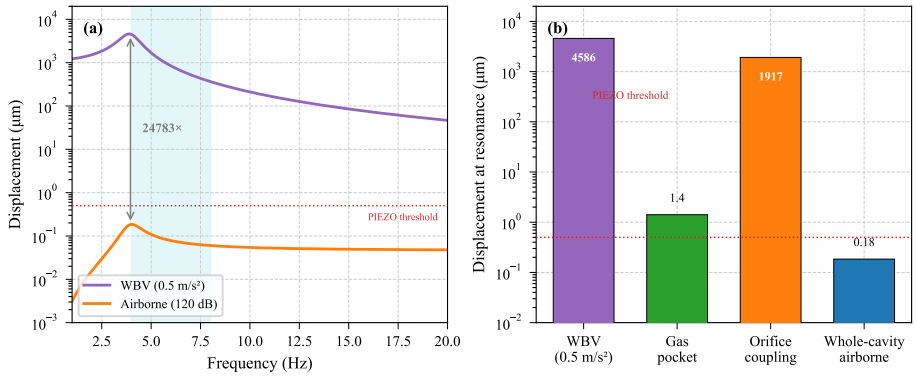

Figure 5: The central result. Left: wall displacement vs. frequency for airborne excitation at 120 dB SPL (lower curve) and mechanical whole-body vibration at $0.5\,\mathrm{m/s^2}$ (upper curve), showing a gap of roughly four orders of magnitude at resonance. Right: comparison of all four coupling pathways at the $n = 2$ resonance; the horizontal dashed line marks the PIEZO cellular mechanosensory reference scale.

5. Discussion

5.1. The Coupling Disparity

Figure 6 summarises this contrast schematically. The central result of this study is the upper-bound coupling ratio 3.3×10^4 separating mechanical from airborne coupling efficiency for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). Including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 1.6 \times 10^4$ (Tables 7–8). The physical origin of this disparity lies in two independent mechanisms:

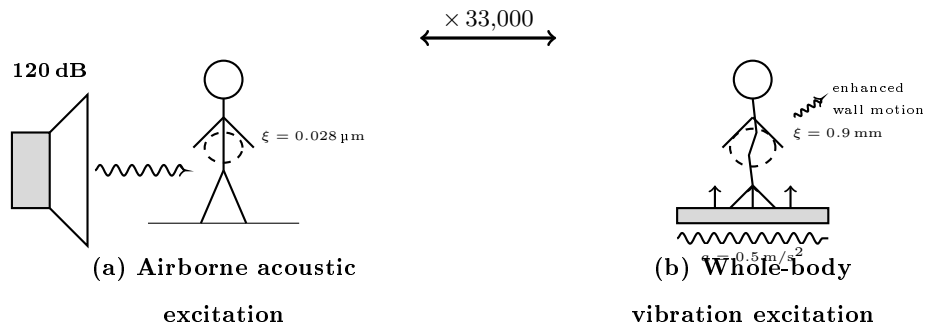

Figure 6: Schematic comparison of the two excitation pathways. (a) Airborne infrasound at 120 dB SPL produces wall displacements of 0.028 µm, well below cellular mechanotransduction benchmarks. (b) Whole-body vibration at occupational levels produces displacements roughly 3.3×10^4 times larger and readily exceeds those benchmarks. Schematic not to scale.

a geometric filtering penalty that suppresses modal-pressure content, and an impedance mismatch that limits energy transfer.

First, the air–tissue impedance mismatch is severe. The specific acoustic impedance of air is $Z_{\text{air}} = \rho_{\text{air}} c_{\text{air}} \approx 420\,\text{Pa}\,\text{s/m}$, while that of soft tissue is $Z_{\text{tissue}} \approx 1.6 \times 10^6\,\text{Pa}\,\text{s/m}$, giving an intensity transmission coefficient $T_I = 4Z_{\text{air}}Z_{\text{tissue}}/(Z_{\text{air}} + Z_{\text{tissue}})^2 \approx 1.1 \times 10^{-3}$. At the body surface, approximately 99.9 % of the incident acoustic power is reflected. This general suppression of transmission across large biological impedance contrasts is consistent with modern frequency-dependent analyses of bone–soft tissue interfaces [39].

Second, even the transmitted pressure must couple to the spatial pattern of a flexural mode. In the long-wavelength (Rayleigh) limit, $ka \ll 1$, the n-th order component of the incident plane wave scales as $(ka)^n$ [18]. At the n=2 resonance ($f_2 \approx 4\,\text{Hz}$) with $R_{\text{eq}} = 0.157\,\text{m}$, we obtain $ka = 2\pi f R/c_{\text{air}} \approx 0.0114$ and therefore $(ka)^2 \approx 1.3 \times 10^{-4}$ for the $n = 2$ flexural mode (equation 16). Physically, the incident wavelength ($\lambda \approx 86\,\text{m}$) is nearly 270 times the body diameter; the pressure field is almost perfectly uniform over the body surface, and the quadrupolar ($n = 2$) pressure gradient that could drive an oblate–prolate deformation is vanishingly small.

The energy budget analysis (§4) confirms this picture from an independent

perspective. The radiation damping ratio in air is $\zeta_{\text{rad}} \approx 10^{-15}$, whereas the structural damping ratio is $\zeta_{\text{struct}} = 0.125$, so the absorption efficiency $\zeta_{\text{rad}}/\zeta_{\text{total}} \sim 10^{-14}$, giving a Breit–Wigner absorption efficiency $4\zeta_{\text{rad}}\zeta_{\text{struct}}/\zeta_{\text{total}}^2 \sim 3 \times 10^{-14}$. The shell is, for practical purposes, acoustically transparent at infrasonic frequencies. By contrast, the upper-bound mechanical model assumes rigid skeletal transmission: the bone–peritoneum impedance contrast is orders of magnitude smaller than the air–tissue mismatch, and the base-excitation drives the $n = 2$ mode via equation 19 with a modal participation factor $\Gamma_2 \approx 0.48$ (§4.3).

This disparity resolves an apparent paradox in the literature. Whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress—including nausea, abdominal pain, and in extreme cases, incontinence [2, 3, 42, 43, 44]—whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [45, 6]. The present modal analysis shows that both observations are consistent with the *same* underlying resonance. The two excitation pathways differ not in the resonance they address, but in the efficiency with which they address it.

5.2. Implications for the Brown Note Hypothesis

The results permit a quantitative evaluation of the "brown note" hypothesis. The hypothesis contains one correct element: abdominal resonance in the 4 Hz to 10 Hz band is physically real, well-supported by the parametric study (§3), and consistent with ISO 2631 transmissibility data [1, 30]. Independent finite-element studies of the seated human body now recover dominant internal and abdominal vibration modes in the 4 Hz to 8 Hz range [46]. The flexural $n = 2$ mode at 4.0 Hz (for $E = 0.1$ MPa, relaxed musculature) falls squarely within the frequency band where occupational exposure studies report gastrointestinal symptoms.

Where the hypothesis fails is in the *excitation pathway.* At 120 dB SPL—a level exceeding the conventional threshold of discomfort and well above any environmental or entertainment exposure—the predicted airborne wall displace-

ment is 0.028 µm (Table 7), roughly two orders of magnitude below the 0.5 µm to 2.0 µm cellular activation range of mechanosensitive PIEZO1 and PIEZO2 ion channels [31, 34, 36] (a benchmark derived from cell-culture experiments; see Limitation 9). In the modern mechanotransduction literature, the more relevant gastrointestinal transducer is probably PIEZO2: somatosensory-neuron PIEZO2 signalling has now been shown to regulate gut transit directly [37]. Reaching the 1 µm cellular benchmark on the energy-consistent basis requires approximately 151 dB (cf. 135 dB for the pressure-based estimate; Table 3), a level associated with severe acoustic hazards and encountered only in explosive blasts and rocket launches. Even at this extreme, the displacement remains three orders of magnitude below what routine whole-body vibration ($a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$) produces (Table 8).

The model also provides the first quantitative explanation for *why* whole-body vibration at 4 Hz to 8 Hz is so effective at producing gastrointestinal effects: it excites a flexural resonance of the fluid-filled abdominal cavity, producing millimetre-scale tissue displacements that far exceed cellular mechanotransduction benchmarks. The epidemiological pattern documented by ISO 2631—peak symptom incidence at 4 Hz to 8 Hz, with a strong dependence on acceleration amplitude—is precisely what a resonance-driven mechanism predicts.

5.3. Alternative Coupling Pathways

The whole-cavity resonance model of §2 treats the abdominal wall as a homogeneous fluid-filled shell. In reality, two additional mechanisms provide alternative pathways for airborne sound to produce tissue displacement: (i) compression of intestinal gas pockets, and (ii) orifice coupling through the upper gastrointestinal tract. We note that the estimates below are preliminary and order-of-magnitude in nature; they are intended to identify plausible mechanisms for future investigation rather than to make quantitative predictions. We consider each in turn.

5.3.1. Intestinal gas pocket response

The human intestine normally contains 100 mL to 200 mL of gas distributed in pockets with effective radii ranging from 1 cm to 3 cm (corresponding to volumes of approximately 4 mL to 110 mL). Unlike the surrounding tissue, these gas volumes are compressible, and they respond to incident pressure by expanding and contracting their bounding walls. If such local deformation is biologically transduced, PIEZO2-dependent gut afferents are the most plausible candidate pathway rather than generic abdominal-wall strain sensing [37].

The classical Minnaert frequency of a spherical gas bubble of radius a in a fluid of density ρ_f is [47]

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}, \tag{27}$$

where $\gamma = 1.4$ is the ratio of specific heats (we use $\gamma = 1.4$ for a diatomic ideal gas; a more representative value for the typical intestinal mixture of $CO_2/CH_4/N_2$ would be $\gamma \approx 1.35$, introducing a $\sim 2\%$ correction to the Minnaert frequency) and P_0 is the ambient pressure. For physiological pocket radii of 1 cm to 3 cm, this yields Minnaert frequencies in the range 110 Hz to 330 Hz—well above the infrasonic band of interest. At 7 Hz, the frequency ratio $f/f_M \approx 0.02$–0.06, so gas pockets are driven far below resonance and experience no resonant amplification ($H \approx 1.0$).

Nevertheless, in the sub-resonant regime the gas pocket wall displacement is governed by the quasi-static compliance of the trapped gas:

$$\xi_{\text{gas}} \approx \frac{p_{\text{inc}}\, a}{3\gamma P_0}, \tag{28}$$

which scales linearly with pocket radius and is independent of frequency. At 120 dB SPL ($p_{\text{inc}} = 20$ Pa) and $a = 2$ cm, this yields $\xi_{\text{gas}} \approx 0.94$ µm—within the PIEZO activation range and roughly 67 times the whole-cavity airborne displacement of 0.028 µm at the same SPL. For larger pockets ($a = 3$ cm), the displacement rises to approximately 1.4 µm, exceeding the lower PIEZO activation threshold.

Crucially, at 7 Hz the acoustic wavelength (49 m) vastly exceeds the body

diameter, so the pressure field inside the body is spatially uniform to within $(ka)^2$ corrections. Although the air–tissue impedance mismatch reflects approximately 99.9 % of incident acoustic intensity ($T_I \approx 10^{-3}$), in the acoustically compact limit ($ka \ll 1$) the body is too small to scatter the incident wave, and the interior pressure equilibrates quasi-statically with the incident field to leading order. Pre-existing gas pockets within the abdomen, presenting a highly compliant inclusion within the otherwise nearly incompressible abdominal fluid, may therefore experience pressure fluctuations comparable to the incident field, potentially driving local volume changes independent of the global cavity resonance. The gas pockets respond to the intracorporeal pressure field without the geometric filtering penalty that suppresses whole-cavity flexural excitation; the mechanism operates as a local pressure-to-displacement transducer embedded within the tissue, rather than as a global shell resonance.

We note speculatively that this gas pocket pathway has two potentially important implications. First, it may explain the well-documented inter-individual variability in infrasound sensitivity: bowel gas content varies enormously (50 mL to 500 mL) with diet, gut motility, and microbiome composition, and individuals with greater gas volume would present more sites for pressure–displacement conversion. Second, the mechanism does not require resonance and operates at any frequency in the sub-wavelength regime, broadening the range of potentially relevant exposures.

5.3.2. Orifice coupling through the upper GI tract

The mouth and nose provide a low-impedance air pathway to the pharynx and—when the esophagus is patent—to the stomach. At 7 Hz, the total path length from mouth to stomach ($L \approx 40$ cm) satisfies $kL \approx 0.05 \ll 1$, so pressure is transmitted through the air column with negligible viscothermal attenuation.

With the mouth open and esophagus patent, the pressure reaching the gastric air bubble is approximately equal to the incident pressure, representing a transmission ratio of order unity compared with the $(ka)^2 \approx 1.3 \times 10^{-4}$ penalty for body-surface coupling—an enhancement factor of roughly 7,700×. Com-

pression of the gastric gas bubble then produces wall displacements by the same mechanism described above for intestinal pockets.

However, this pathway is severely limited by anatomy. The esophagus is a collapsible muscular tube that is closed at rest; it opens only during swallowing (approximately 0.5 s to 1.0 s per event), belching, or pathological gastroesophageal reflux. At rest with the mouth open, the air column terminates at the upper esophageal sphincter, and pressure reaches only the supraglottic space. The orifice pathway is therefore *intermittent*, dependent on the state of the upper GI sphincters, and cannot be relied upon as a continuous coupling mechanism.

5.3.3. Assessment of alternative pathways

Table 9 summarises the predicted displacement for each coupling mechanism at 120 dB SPL and 7 Hz.

Table 9: Comparison of coupling mechanisms at 120 dB SPL, 7 Hz.

Mechanism	ξ (μm)	Exceeds 1 µm?
Cavity resonance (n=2, airborne)	0.028	No
Gas pocket $a = 2$ cm (airborne)	0.94	Marginal
Gas pocket $a = 3$ cm (airborne)	1.4	Yes
Orifice (open mouth $\rightarrow$ stomach)	$\sim$0.5	Marginal
WBV at 0.5 m/s^2 (mech.)	4,586	Yes
WBV at 1.15 m/s^2 (mech.)	10,549	Yes

Even with the most favourable alternative pathway (a 3 cm-radius gas pocket at 120 dB), the airborne displacement remains two to three orders of magnitude below what mechanical coupling achieves at the EU Whole-Body Vibration Directive action value of 0.5 m/s^2. The alternative pathways therefore do not alter the principal conclusion of this work—that airborne infrasound is an ineffective excitation route compared with mechanical vibration—but they provide a physically grounded explanation for why rare, sporadic gastrointestinal responses

to extreme infrasound (>120 dB) are not entirely implausible, and why such responses exhibit large inter-individual variability.

5.4. Clinical and Occupational Relevance

The coupling disparity has direct implications for occupational exposure standards. ISO 2631-1 specifies frequency-weighted acceleration limits for whole-body vibration, with the highest sensitivity weighting in the 4 Hz to 8 Hz band—precisely the range where our model predicts the $n = 2$ flexural resonance [1], and consistent with von Gierke and Brammer's compilation of organ resonance data [11]. More recent occupational reviews continue to link whole-body vibration exposure with adverse health outcomes across multiple organ systems [44]. The EU Physical Agents (Vibration) Directive 2002/44/EC codifies two thresholds: a daily exposure action value of $0.5\,\mathrm{m/s^2}$ $A(8)$ and an exposure limit value of $1.15\,\mathrm{m/s^2}$ $A(8)$. (The $A(8)$ action and limit values are frequency-weighted eight-hour equivalent accelerations; we use them here as illustrative equivalent harmonic amplitudes at the n=2 resonance frequency, not as direct predictions under the Directive's time-weighted assessment protocol.) At the action value, our model predicts a relative abdominal wall displacement of 4.6 mm at resonance (Table 8)—more than 4,000× the PIEZO cellular activation benchmark. The present analysis therefore provides a *mechanistic rationale* for why these particular frequency bands and amplitude thresholds were empirically selected: they correspond to conditions under which flexural resonance of the abdominal cavity produces tissue strains large enough to activate mechanotransduction pathways.

For airborne infrasound, ISO 7196 defines G-weighting but prescribes no dose–response limits. Our analysis supports this regulatory asymmetry: at environmental and occupational SPL (≤ 120 dB), airborne coupling to abdominal flexural modes is negligible, and exposure limits based on abdominal resonance effects are unnecessary at these levels.

A potentially valuable extension of this framework is toward personalised vibration exposure assessment. The parametric study (§3) shows that the $n = 2$

eigenfrequency depends sensitively on body habitus: larger abdominal radius (higher BMI) reduces the resonance frequency, while greater muscular tone (higher effective E) increases it. In principle, imaging-derived geometric and viscoelastic parameters [27] combined with the analytical expressions of §2 could yield subject-specific resonance frequency estimates, enabling targeted vibration exposure limits for workers whose body habitus places their resonance frequency in a particularly hazardous band.

5.5. Broader Applications of the Analytical Framework

The brown note provided the motivation for this work, but the analytical framework—a fluid-filled viscoelastic shell, a modal decomposition into breathing and flexural families, and a systematic comparison of airborne versus mechanical coupling efficiency—is considerably more general. Any biological or engineered structure that can be idealised as a compliant enclosure surrounding a dense fluid falls within the scope of the model developed in §2. The two key dimensionless groups that govern the physics, the Helmholtz number ka and the impedance ratio $Z_{\text{wall}}/Z_{\text{fluid}}$, transfer directly; what changes between applications is the frequency range, the material properties, and the geometric aspect ratio of the shell. Potential applications include primary blast injury modelling—where ka approaches unity and the geometric filtering penalty largely disappears [48, 49]—swim-bladder target-strength prediction in fisheries acoustics [50, 51], and therapeutic ultrasound dosimetry where coupling must be maximised rather than shown to be negligible [52]. These extensions are beyond the scope of the present work but follow directly from the same modal decomposition and coupling-comparison methodology.

5.6. Nonlinear Amplitude Effects

The linear model of §2 is valid only for small displacements relative to the wall thickness. At the large amplitudes associated with whole-body vibration near the EU exposure limit, geometric nonlinearity from membrane stretching and fluid volume conservation may stiffen the response, further reducing actual

displacement below the linear prediction. Such hardening behaviour is characteristic of thin-shell vibration at moderate amplitudes [23] and would shift the resonance peak to higher frequency while reducing its amplitude—effects that are conservative for the brown note argument, since they widen the gap between prediction and any putative airborne effect. The airborne displacement ($\sim 0.028\,\mu\mathrm{m}$) remains deep in the linear regime regardless. A detailed nonlinear analysis, including the amplitude-dependent frequency shift and hysteretic jump phenomena expected at occupational vibration levels, is deferred to future work.

5.7. Limitations

We identify the following limitations, grouped by their likely effect on the conclusions.

1. **Linear theory.** The Love–Kirchhoff shell formulation assumes small displacements. As discussed in §5.6, occupational vibration amplitudes likely exceed the linear regime, and the resulting hardening nonlinearity would reduce peak displacement below the linear prediction. Because the airborne displacement remains deep in the linear regime, nonlinearity *reduces* the coupling ratio somewhat but does not alter the qualitative conclusion.
2. **Homogeneous, isotropic wall.** The abdominal wall is a layered anisotropic composite comprising peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. The homogenisation error in effective bending stiffness is expected to be modest compared with inter-subject variability in wall thickness and elastic modulus. The isotropic assumption is a first-order approximation; anisotropy would split the mode degeneracies of the equivalent sphere but would not change the order of magnitude of the eigenfrequencies or the coupling ratio. Recent abdominal-wall mechanics reviews and elastography surveys place the canonical $E = 0.1\,\mathrm{MPa}$ and $\eta = 0.25$ values as representative of contemporary measurement ranges, so the remaining uncertainty is quantitative rather than order-of-magnitude [24, 27].

3. **Free-shell boundary conditions.** The abdomen is partially constrained by the spine posteriorly, the pelvis inferiorly, and the rib cage superiorly. The Rayleigh–Ritz study (§3.5) is best understood as an *exploratory sensitivity study*: it shows the range of possible frequencies under progressively constrained assumptions, from a free sphere ($f_2 = 4.13\,\mathrm{Hz}$) to 25% surface clamped ($f_2 = 3.66\,\mathrm{Hz}$, -11.4%). That clamping *lowers* f_2 is a consequence of the fluid-dominated inertia: the added mass ($7.4\times$ the shell mass for $n = 2$) means that constraining the wall reduces the effective modal stiffness more than it reduces the kinetic energy. The entire bracketed range lies well within the material-property uncertainty band, and the coupling ratio is unaffected because both excitation pathways address the same modal structure.
4. **Thin-shell approximation.** The wall-thickness-to-radius ratio ($h/R \approx 0.06$) lies near the conventional upper bound of thin-shell theory ($h/R < 0.05$). First-order shear-deformation (Mindlin–Reissner) corrections would reduce the $n = 2$ frequency by approximately 4 % [53], a negligible effect compared with the material-property uncertainty. This limitation is unlikely to affect any conclusions.
5. **Equivalent-sphere geometry.** The oblate spheroidal abdomen is mapped to an equivalent sphere via $R_{\mathrm{eq}} = (a^2c)^{1/3}$. This preserves volume and leading-order curvature but suppresses the equatorial–polar mode splitting that occurs in an oblate spheroid [54]. For a typical aspect ratio $c/a = 0.67$, the splitting is estimated at $\sim$33 %, which could shift individual mode frequencies but would not change the coupling ratio or the qualitative separation between breathing and flexural mode families.

   More fundamentally, the human abdominal cavity is a *triaxial* ellipsoid (lateral $\sim$14 cm, craniocaudal $\sim$16 cm, anteroposterior $\sim$10 cm), not a spheroid of either kind. The oblate approximation captures the principal feature—anterior–posterior flattening—but any spheroid is a simplification. Crucially, however, the equivalent-sphere flexural frequencies depend on R_{eq} alone: under the equivalent-sphere formula, at fixed R_{eq}, varying c/a from

0.3 (extremely oblate) to 3.0 (extremely prolate) changes f_2 by less than 1 %. The Ritz oblate-spheroid correction introduces additional aspect-ratio dependence (see Section 2), but the geometry enters the coupling ratio primarily through R_{eq}, so the qualitative results are insensitive to the choice of spheroidal idealisation.

6. **Inviscid fluid assumption.** The enclosed fluid is modelled as inviscid. A viscous correction using the Stokes boundary-layer thickness shows that even at the worst-case mucus viscosity ($\mu = 0.1\,\mathrm{Pa\,s}$), the quality factor decreases by only $\Delta Q = -1.85\%$. The crossover viscosity at which viscous damping would equal structural damping is $281\,\mathrm{Pa\,s}$, far above physiologically realistic abdominal fluid viscosities. The inviscid assumption is therefore well justified for any physiologically realistic abdominal fluid.
7. **No internal organ structure.** Visceral organs are homogenised into a single fluid. This estimate follows from a Hashin–Shtrikman upper-bound effective-medium calculation for a fluid matrix containing soft solid inclusions [55]: taking $K_{\mathrm{tissue}} \approx K_{\mathrm{water}}$, $\mu_{\mathrm{tissue}} \approx 1\,\mathrm{kPa}$, and a volume fraction $\phi = 0.25$ (representative of liver, spleen, and kidneys combined) predicts a frequency shift of only 0.4 % for the $n = 2$ mode. Because the organ tissue bulk modulus is close to that of water, the effective compressibility barely changes, and at sub-percolation volume fractions the isolated inclusions cannot transmit a system-spanning shear network. The homogeneous fluid model is therefore adequate for the present purposes.
8. **Gas pocket model simplifications.** The gas pocket analysis (§5.3) treats intestinal gas as spherical bubbles in an infinite homogeneous fluid, neglecting the cylindrical geometry of the gut lumen, viscoelastic wall loading, and the surrounding organ structure. These simplifications likely *overestimate* gas pocket displacement, because intestinal wall mass loading and stiffness would reduce the effective compliance. A coupled gas–tissue finite-element model is needed to refine these estimates.
9. **Mechanotransduction benchmarks.** The 0.5 µm to 2.0 µm PIEZO activation range [31, 34, 36] derives from cell-culture and membrane-tension

experiments and may not directly translate to tissue-scale effects, where strain concentrations around cells are modulated by the extracellular matrix. We therefore treat these values as cellular mechanosensory reference scales rather than experimentally confirmed tissue-scale thresholds. In gastrointestinal tissue the relevant transduction pathway is also likely to involve PIEZO2-expressing afferents rather than PIEZO1 alone [37]. This limitation introduces uncertainty in the absolute benchmark but does not affect the *relative* comparison between airborne and mechanical coupling.

In summary, the identified limitations affect the absolute values of predicted frequencies and displacements—in several cases substantially—but none alter the central finding that airborne acoustic coupling to abdominal flexural modes is 10^4–10^5 times weaker than mechanical coupling under occupationally relevant conditions.

5.8. Proposed Experimental Validation

We propose a two-stage experimental programme to validate the key predictions of the analytical model. The first stage employs a tissue-mimicking phantom: an oblate spheroidal silicone shell ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$) filled with degassed water, matching the canonical geometry of §2. The shell material is platinum-cure silicone (Ecoflex 00-30 to Dragon Skin 10A), whose Young's modulus spans 0.07 MPa to 0.15 MPa—bracketing the relaxed abdominal wall estimate of $E = 0.1\,\mathrm{MPa}$. By casting phantoms at several mixing ratios, the $f_2 \propto \sqrt{E}$ dependence can be tested directly. Modal frequencies are measurable with an instrumented impact hammer or a low-frequency shaker sweep, and mode shapes can be reconstructed with a scanning laser vibrometer (Polytec PSV-500). The predicted $n = 2$ frequency for the recommended Ecoflex 00-30 formulation is 2.5 Hz; for Dragon Skin 10A, 3.6 Hz; the model predicts 4.0 Hz for the canonical human case.

The critical measurement is the coupling-ratio comparison. By mounting the phantom on an electrodynamic shaker (mechanical pathway) and separately

exposing it to a subwoofer-generated infrasound field at 120 dB (airborne pathway), the displacement ratio $\xi_{\mathrm{mech}}/\xi_{\mathrm{air}}$ can be measured in a single experimental session. The model predicts ratios of order 10^4–10^5 across the silicone formulation range (Table 9). Non-detection of the airborne response at the vibrometer noise floor ($\sim 0.01\,\mu\mathrm{m}$) would itself constitute a positive validation of the weak-coupling prediction. The total cost of consumables (silicone, molds, sensors) is estimated at less than US$ 3 100, assuming access to shared laboratory equipment (shaker, laser vibrometer, DAQ).

The second stage extends to in vivo human subjects. Subject-specific abdominal wall moduli would be obtained via magnetic resonance elastography or shear-wave ultrasound elastography, and whole-body vibration transmissibility measured on an ISO 2631-compliant platform at sub-action-value amplitudes ($0.3\,\mathrm{m/s^2}$). A sample of 15 subjects, spanning a range of body habitus, is sufficient (power $= 0.80$, $\alpha = 0.05$) to detect a paired-test effect size $d = 1.0$ between measured and predicted peak-transmissibility frequencies. This programme requires Institutional Review Board approval and is estimated at 8–12 months, with the phantom study serving as a prerequisite validation of the model physics under controlled conditions.

6. Conclusions

The central result of this study is an upper-bound coupling disparity of approximately 3.3×10^4 between mechanical whole-body vibration and airborne acoustic excitation of the human abdominal cavity for 120 dB SPL airborne loading and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). This single quantity, derived from first-principles shell theory and energy-consistent reciprocity analysis, resolves a long-standing empirical asymmetry: occupational whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects, while airborne infrasound at comparable frequencies and extreme intensities does not.

The disparity has a clean physical origin. The abdominal cavity possesses

two distinct mode families: a breathing mode ($n = 0$) near 2500 Hz, governed by the fluid bulk modulus and therefore irrelevant at infrasonic frequencies; and flexural modes ($n \geq 2$) in the 4 Hz to 10 Hz range that match ISO 2631 abdominal transmissibility data. Airborne sound at these frequencies has a wavelength exceeding 85 m, placing the abdomen deep in the Rayleigh regime ($ka \approx 0.01$). The $(ka)^n$ geometric penalty suppresses modal-pressure content by four orders of magnitude; compounding this, the extremely low radiation efficiency at the air–tissue interface limits the energy available to drive the global resonance. An energy-consistent analysis confirms an absorption efficiency of $\sim 3 \times 10^{-14}$: the shell is, for practical purposes, acoustically transparent. Reaching the cellular mechanotransduction benchmark via airborne infrasound alone would require approximately 151 dB on the energy-consistent basis (135 dB using the less physical pressure-based estimate)—a level at which structural damage to the middle ear and other severe acoustic hazards would present more immediate concerns than any gastrointestinal effects.

Mechanical vibration, transmitted through the skeleton and abdominal wall, couples directly and, for the headline comparison (120 dB SPL airborne versus $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS WBV with $\Gamma_2 = 1$), produces displacements approximately 3.3×10^4 times larger; including the participation correction leaves $\sim 1.6 \times 10^4$. This disparity is not a subtle biomechanical effect but a straightforward consequence of long-wavelength acoustics and impedance physics. Monte Carlo uncertainty quantification using the pressure-based airborne upper bound yields a ratio spanning roughly one and a half orders of magnitude, with a 90% credible interval of $[2 \times 10^3,\ 3 \times 10^4]$; elastic modulus accounts for 86% of total variance.

The underlying resonance is genuine, but the airborne-coupling mechanism commonly invoked in brown-note claims is not supported. What popular culture attributes to airborne sound is more consistently explained by mechanical vibration—a distinction that matters considerably for occupational health standards. Airborne infrasound cannot excite the global cavity resonance to physiologically relevant amplitudes; whether localised compressible inclusions

(intestinal gas pockets) provide a secondary transduction pathway at extreme intensities remains an open question, as they operate by quasi-static compliance rather than cavity resonance and are explored separately.

Future work should verify the boundary condition assumptions through finite element analysis of anatomically realistic geometries, investigate the role of bowel gas pockets as localised acoustic transducers, and extend the model into the nonlinear regime relevant to high-amplitude mechanical excitation.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Formal Analysis, Software, Writing – Original Draft, Writing – Review & Editing. **Brian R. Mace**: Methodology, Validation, Writing – Review & Editing.

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During the preparation of this work the authors used GitHub Copilot (GitHub, Inc.; powered by large language models from OpenAI and Anthropic) in order to assist with: (1) implementing analytical expressions as Python code for numerical evaluation, (2) generating publication figures from computed results, (3) conducting structured literature searches, and (4) drafting and revising manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.